

MIA MORRISON



Website

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PROFESSIONAL SUMMARY

Aerospace Engineering student focused on guidance, navigation, and control (GNC), with hands-on experience in MATLAB, Python, simulation, robotics, and mechanical design. Developing experience in spacecraft GNC through senior design and technical projects. Strong background in leadership, communication, and collaborative problem-solving.

EDUCATION

EMBRY-RIDDLE AERONAUTICAL UNIVERSITY (ERAU) | B.S. Aerospace Engineering | GPA: 3.33

Est. May 2027

ENGINEERING EXPERIENCE

GNC ENGINEER— LUNAR CUBESAT SENIOR DESIGN

Sep 2026 – May 2027

- Responsible for GNC design and analysis for a lunar communications-relay CubeSat, including orbital mechanics, attitude dynamics, and spacecraft stabilization.
- Support development of trajectory and feedback-control concepts to meet mission and spacecraft requirements.

DOCKING ROBOTICS | MECHANICAL ENGINEERING INTERN | SolidWorks

May 2025 – Sept 2025

- Designed and 3D-printed a modular drone landing dock in SolidWorks using a six-panel interlocking system optimized for strength, precision, and rapid assembly.

TECHNICAL PROJECTS

PATHFINDER AODR-01 AUTONOMOUS ROVER | C++(Arduino), Onshape, 3D Printing, PCBWay Sponsorship

June 2026 – Present

- Independently designed and built an autonomous obstacle-detection rover using an Arduino Nano, ultrasonic sensor, DC motors, LCD display, and custom 3D-printed chassis, with self-taught Arduino programming and embedded systems.
- Selected for a PCBWay manufacturing sponsorship; redesigned prototype currently in professional production.**

SIMULINK 3D ROCKET LAUNCH SIMULATION | MATLAB, Simulink

July 2026

- Developed a 3D rocket flight simulation modeling variable-mass dynamics, thrust, drag, gravity, and translational motion in MATLAB and Simulink.
- Implemented 3-2-1 Euler attitude kinematics, body-to-inertial force transformations, and a custom MATLAB trajectory animation with real-time trajectory and flight telemetry updates.

PYTHON Satellite Trajectory Analysis Tool | Python: Tkinter, NumPy, Matplotlib, PyAstronomy, SGP4

July 2026

- Developed an interactive desktop application for satellite trajectory analysis and orbit visualization using Python.
- Implemented real-time 3D orbit visualization through adjustable Keplerian orbital elements and generated altitude profiles from TLE data using the SGP4 propagator. Integrated data visualization, GUI development, and CSV export.

CATIA SCOOTER DESIGN

Sept 2023 – Dec 2023

- Reverse-engineered and modeled a complete electric scooter assembly using CATIA.
 - Created detailed engineering drawings, technical documentation, and assembly models.
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Software Skills: Simulink, MATLAB, Python, C++(Arduino), Linux, SolidWorks, Onshape, CATIA, Microsoft 365

Leadership Skills: Public Speaking, Project Management, Team Leadership, Youth Mentorship, Community Engagement

LEADERSHIP & COMMUNITY EXPERIENCE

NATIONAL PUBLIC SPEAKER | YOUTH ADVOCATE (Boys & Girls Clubs of America)

June 2022- Present

- Selected as the 2022 & 2023 Florida Youth of the Year; Advocated for mental health awareness and educational resources.
- Created and led a college readiness program mentoring students through applications, scholarships, and essays; delivered keynote presentations at national conferences, including NASCAR Business.